

# Week 7 Workshop: Part 1

Recall in our Week 6 Tutorial, we estimate the multi-factor models.

```
library(frenchdata)
```

Warning: package 'frenchdata' was built under R version 4.5.2

```
#####  
# (a) "Fama/French 3 Factors" Monthly  
#####  
  
ff3 <- download_french_data("Fama/French 3 Factors")  
  
ff3_monthly_raw <- ff3$subsets$data[[1]]  
  
# convert to xts object  
library(xts)
```

Warning: package 'xts' was built under R version 4.5.2

Warning: package 'zoo' was built under R version 4.5.2

```
# if date is like 192607, 202312, etc.  
ff3_monthly_raw$date <- as.yearmon(as.character(ff3_monthly_raw$date), format = "%Y%m")  
ff3.xts <- xts(  
  ff3_monthly_raw[, c("Mkt-RF", "SMB", "HML", "RF")] , # drop the date column  
  order.by = as.Date(ff3_monthly_raw$date)           # month index  
)  
  
# Optional: Rename the columns  
colnames(ff3.xts) <- c("mkt_excess", "smb", "hml", "rf")  
  
# convert from "1926-07-01" to "Jul 1926"  
index(ff3.xts) <- as.yearmon(index(ff3.xts))
```

```
#####
# (b) "5 Industry Portfolios"
#####

# Download the dataset (exact name on Ken French's library)
ind5 <- download_french_data("5 Industry Portfolios")

# Grab the "Average Value Weighted Returns -- Monthly"
id5.df <- ind5$subsets$data[[1]]

# convert to xts
id5.df$date <- as.yearmon(as.character(id5.df$date), format = "%Y%m")
id5.xts <- xts(
  id5.df[, c("Cnsmr", "Manuf", "HiTec", "Hlth", "Other")] , # drop the date column
  order.by = as.Date(id5.df$date) # month index
)

# convert from "1926-07-01" to "Jul 1926"
index(id5.xts) <- as.yearmon(index(id5.xts)) # match by year-month

#####
# (c) Momentum Factor (Mom)
#####

# 1) Download the dataset (exact name on Ken French's library)
mom_raw <- download_french_data("Momentum Factor (Mom)")

# 2) Grab [[1]] monthly
mom_raw.df <- mom_raw$subsets$data[[1]]

# convert to xts
mom_raw.df$date <- as.yearmon(as.character(mom_raw.df$date), format = "%Y%m")
mom.xts <- xts(
  mom_raw.df[, "Mom" ], # drop the date column
  order.by = as.Date(mom_raw.df$date) # month index
)

# convert from "1926-07-01" to "Jul 1926"
index(mom.xts) <- as.yearmon(index(mom.xts)) # match by year-month

#####
# January 1997 to December 2025

id5.xts <- id5.xts["1997-01/2025-12"]
```

```
ff3.xts <- ff3.xts["1997-01/2025-12"]

mom.xts <- mom.xts["1997-01/2025-12"]

# unrestricted multi-factor
ff4f.fit <- lm(id5.xts$Cnsmr~ff3.xts$rfrff3.xts$mkt_excess+ff3.xts$smb+ff3.xts$hml+mom.xts
summary(ff4f.fit)
```

Call:

```
lm(formula = id5.xts$Cnsmr - ff3.xts$rfrff3.xts$mkt_excess +
    ff3.xts$smb + ff3.xts$hml + mom.xts)
```

Residuals:

|  | Min     | 1Q      | Median | 3Q     | Max    |
|--|---------|---------|--------|--------|--------|
|  | -7.9443 | -1.1321 | 0.0826 | 1.0188 | 6.3725 |

Coefficients:

|                     | Estimate | Std. Error | t value | Pr(> t )    |
|---------------------|----------|------------|---------|-------------|
| (Intercept)         | 0.15107  | 0.10602    | 1.425   | 0.15508     |
| ff3.xts\$mkt_excess | 0.84740  | 0.02488    | 34.060  | < 2e-16 *** |
| ff3.xts\$smb        | -0.09592 | 0.03351    | -2.862  | 0.00446 **  |
| ff3.xts\$hml        | 0.07909  | 0.03192    | 2.477   | 0.01372 *   |
| mom.xts             | -0.01853 | 0.02270    | -0.816  | 0.41490     |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.933 on 343 degrees of freedom

Multiple R-squared: 0.8, Adjusted R-squared: 0.7976

F-statistic: 343 on 4 and 343 DF, p-value: < 2.2e-16

Following from Week 6 Tutorial, answer the following questions.

6. What can we conclude about the OLS standard errors?

The presence of serial correlation, and heteroskedasticity means that the OLS standard errors are incorrect. Thus all hypothesis tests are invalid (since our regression standard errors are incorrect). However, the OLS estimates are still unbiased.

7. How would you fix this issue?

Serial correlation in the error means that there is unidentified time-dependence in the series. So it is best to try and identify the lag structure that is present, and then incorporate it into our regression. In this case, we do not need to do anything.

To fix serial correlation and heteroskedasticity, we will use HAC standard errors. These are valid in large samples (which we generally have with financial data). We estimate a new equation with the same variables, but we use asymptotically valid standard errors.

Run the code below.

```
# HAC

# need both of the following libraries
library(lmtest)
```

Warning: package 'lmtest' was built under R version 4.5.2

```
library(sandwich)
```

Warning: package 'sandwich' was built under R version 4.5.2

```
T <- nobs(ff4f.fit)
m <- floor(0.75 * T^(1/3))

# compute newey-west HAC uses library(sandwich)
NW_VCOV <- NeweyWest(ff4f.fit,
                      lag = m - 1, prewhite = F,
                      adjust = T)

# Std errors, t-test, p-value uses library(lmtest)
coeftest(ff4f.fit, vcov = NW_VCOV)
```

t test of coefficients:

|                     | Estimate  | Std. Error | t value | Pr(> t )   |
|---------------------|-----------|------------|---------|------------|
| (Intercept)         | 0.151071  | 0.107181   | 1.4095  | 0.1596     |
| ff3.xts\$mkt_excess | 0.847402  | 0.033440   | 25.3406 | <2e-16 *** |
| ff3.xts\$smb        | -0.095924 | 0.064005   | -1.4987 | 0.1349     |
| ff3.xts\$hml        | 0.079087  | 0.062110   | 1.2733  | 0.2038     |
| mom.xts             | -0.018530 | 0.033886   | -0.5468 | 0.5848     |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

8. After fixing this issue, go back and test whether the SMB factor is statistically significant.

Note that the coefficient estimates have not changed, but the standard errors have. Generally HAC standard errors will be larger than OLS standard errors, however this is not always the case in practice.

The standard error for  $\hat{\beta}_{SMB}$  has now increased to 0.064 from 0.034 using OLS. Consequently, the p-value has also increased to 0.1349 from 0.0045, so we would now change our conclusion. The SMB variable is not statistically significant once the other variables are accounted for, suggesting that the consumer portfolio does not behave like a big-stock portfolio (or a small-stock portfolio).